

Rishon Jos Anton S

AI & Machine Learning Engineer | LLM Systems & Research

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Education

St. Joseph's College of Engineering

Chennai, India

B.Tech in Artificial Intelligence and Machine Learning — **CGPA: 9.01/10**

Sep 2023 – Present

– **Relevant Coursework:** Data Structures & Algorithms, DBMS, Deep Learning, Natural Language Processing

Experience

Qik Enterprises Private Limited

Chennai, India

Software Developer Intern (LLM Systems)

Jun 2025 – Dec 2025

- **Cross-Platform Integration:** Engineered a Manifest V3 Chrome Extension and backend service integrating an **LLM meeting intelligence platform** across Google Meet, Zoom, and Microsoft Teams.
- **Real-Time Streaming:** Built a dual audio capture pipeline (`tabCapture` & `getUserMedia`) streaming to a backend **Whisper ASR & LLM pipeline** (<2s latency), reducing manual note-taking overhead by 65%.
- **Transcript RAG:** Integrated **Retrieval-Augmented Generation (RAG)** over transcripts using vector embeddings, enabling semantic search and instant natural language QA across meeting logs.

Projects

DocuMind-RAG: Advanced Hybrid & Agentic RAG Engine

June 2026 – July 2026

Tech Stack: PyTorch, LangGraph, BM25, Cross-Encoder Reranking, HyDE, Ollama Llama 3.2

- **Hybrid Retrieval:** Architected a production RAG pipeline combining Parent-Child Chunking, Native NumPy Vector Indexing, and BM25 + Dense Hybrid Search via RRF, achieving a **+34% MRR@5 gain** over naive vector search.
- **Re-ranking & Expansion:** Integrated Cross-Encoder Re-ranking (`ms-marco-MiniLM`) and HyDE (Hypothetical Document Embeddings) to maximize retrieval precision at **<150ms query latency**.
- **Agentic State Graph:** Implemented a LangGraph Self-RAG agent executing automated document relevance grading, dynamic query rewriting, and hallucination-free context generation.

AttentionBench: NMT Compositional Extrapolation & Interpretability

March 2026 – April 2026

Tech Stack: PyTorch, Attention (Bahdanau/Luong/RoPE), Mechanistic Interpretability, SciPy

- **Extrapolation Benchmark:** Evaluated 5 PyTorch Seq2Seq/Transformer architectures (Vanilla GRU, Bahdanau, Luong, Trans-Abs, Trans-RoPE) on zero-shot compositional length extrapolation ($\sim 800K$ params).
- **Mechanistic Diagnostics:** Developed a grammar-aware Failure Taxonomy and teacher-forced Oracle Logit & Cross-Attention Entropy diagnostics ($entropy \leq 0.12nats$), isolating decoder premature `<eos>` early-stopping bias.

NanoGPT: Custom Decoder-Only GPT & Tokenizer from Scratch

Jan 2026 – Feb 2026

Tech Stack: PyTorch, Python, CUDA, Byte-Pair Encoding (BPE), Autoregressive Decoding

- **Custom Transformer:** Engineered a decoder-only GPT model from scratch in PyTorch featuring Causal Self-Attention, RoPE, KV-caching (**3.2x faster inference**), and a custom BPE subword tokenizer.
- **Decoding Optimization:** Evaluated autoregressive decoding strategies, achieving competitive validation loss (**perplexity 14.2**) via Temperature, Top-K, and Nucleus (Top-P) sampling.

Technical Skills

Languages: Python, C, C++

LLM & GenAI: PyTorch, Transformers, RAG, Hybrid Search (BM25 + Dense), LangGraph, Vector Stores, Cross-Encoders, Prompt Engineering

Core ML & Frameworks: Mechanistic Interpretability, Logit Lens, Cross-Attention Diagnostics, Scikit-learn, SciPy, NumPy, FastAPI, PyTorch

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS

Achievements

- **SAP Hackfest 2025 Finalist** — Selected among the Top 70 out of 5,200+ teams nationally.
- **Datathon 3.0 National Hackathon Finalist** — Ranked in the Top 8 out of 300+ competing teams.
- **ProjectFest Exhibition Winner** — Ranked Top 5 out of 110 teams; awarded Rs. 10,000 prize.
- **Maria Philip Debate 2025–26 State-Level Runner-Up** — Awarded Rs. 15,000 prize.
- **ICT YouthTalk 2025–26 Regional Finalist** — Ranked in Top 8 among 500+ participants.